

Report for Summer Analytics Hackathon (Week 2)

Problem Statement

This task was to classify into two classes where one needs to decide if an online customer will convert (make a purchase) or not, using 12 behavioral and demographic attributes. The performance criterion will be the F1-Score evaluated using a secret private test set where both precision and recall are considered to be important due to the class imbalance issue (31% conversions).

Dataset

- **train.csv** – 10,000 rows with labels
- **public_test.csv** – 3,000 rows with labels (which was also used as training dataset to extend the training data)
- **private_test.csv** – 3,000 rows without labels (final prediction file)

As the public_test.csv file has the Converted label, it was combined with the train.csv file to form a new training file with 13,000 rows.

Preprocessing

- **Missing Values Handling:** Missing values were handled by Median imputations (resistant to outliers). The Imputer was trained using the training dataset alone, to avoid data leakage.
- **Feature Transformation:** LabelEncoder transformations were applied to the columns 'Device_Type' and 'Traffic_Source'.
- **Feature Scaling:** StandardScaler (mean = 0, sd = 1) used post-imputation. Necessary for convergence of Logistic Regression model and fair coefficients comparison.

All transformations applied to the training set and then to private test data to ensure no data leakage.

Feature Engineering

12 new features were derived from existing columns to provide richer signals:

New Feature	Formula	Intuition
Engagement_Score	Pages x Products x (Time + 1)	Session intensity
Products_Per_Page	Products / (Pages + 1)	Focus - How targeted is the user
Time_Per_Page	Time / (Pages + 1)	How much attention was devoted to each page
Is_Returning	1 if Previous_Purchases > 0	Loyal Customer - Conversion indicator
Discount_Loyal	Discount_Seen x Is_Returning	Returning customer with Discount seen
Income_Per_Age	Income / (Age + 1)	Purchasing power relative to age
Log_Income	$\log(1 + \text{Income})$	Normalizes Right Skewed Income

Log_Time	$\log(1 + \text{Time_On_Site})$	Normalizes Right Skewed Session Time
Log_Engagement	$\log(1 + \text{Engagement_Score})$	Normalizes Right Skewed Engagement
Pages_x_Products	Pages x Products	Deep Browsing
Discount_x_Pages	Discount_Seen x Pages_Viewed	Deep Browsing With Discount Seen
Previous_x_Discount	Previous_Purchases x Discount_Seen	Strong Loyalty And Discount Signal

Model : Logistic Regression

The Logistic Regression model computes the probability of conversion through the use of a sigmoid function on a linear combination of the features. The following parameters are defined:

class_weight='balanced' : uses automatic weights for minority class since we have an imbalance of 31/69.

C=0.1 : more L2 regularization for better fit.

max_iter = 1000 : enough number of iterations for the solver.

Validation Strategy

I have used Stratified 5-fold cross-validation to ensure that each fold maintains the same class distribution as the initial dataset. For each fold:

1. Trained the model on 4 folds and tested on one fold.
2. Got predicted probabilities for the hold-out fold.
3. Optimized the threshold. Thresholds were chosen between 0.20 and 0.80 in steps of 0.01. The threshold which maximizes F1 metric on the validation fold is stored.

The threshold for prediction on the test dataset was calculated as the average of all fold thresholds (~0.43).

Results

Mean CV F1 Score: ~0.565 (compared to ~0.45 from starter notebook baseline)

Contributing factors to better results compared to baseline:

- a) Combining of public_test with training dataset
- b) 12 new features created
- c) Log transformations
- d) Class balancing
- e) Threshold optimization

Methodology Summary Table

Problem	Solution	Significance
Missing Values	Median Imputation	Robust to outliers

Categoricals	Label Encoding	Simple, effective for tree-like splits
Feature Engineering	12 new features	Captures interaction & behavioral signals
Scaling	StandardScaler	Required for Logistic Regression
Model	Logistic Regression	Principled classifier; course syllabus
Imbalance	class_weight balanced	Prevents majority-class bias
Validation	Stratified 5-Fold CV	Preserves class ratio across folds
Threshold	F1-tuned per fold	0.5 is suboptimal for F1 metric